

43. Jaclyn, an investor, begins a start-up to revitalize homes in South Bend, Indiana. She begins with \$10,000, making her investor 1. Each investor that joins will invest \$500 more than the previous investor. How much does the 50th investor invest in the project? With that 50th investor, what is the total amount invested in the project?
44. Jasmyn has decided to train for a marathon. In week one, Jasmyn runs 5 miles. Each week, Jasmyn increased the running distance by 2 miles. After training for 14 weeks, how many total miles will Jasmyn have run?
45. The base of a pyramidal structure has 144 blocks. Each level above has 5 fewer blocks than the previous level. How many total blocks are there if the pyramidal structure has 25 levels?
46. As part of a deal, a friend tells you they will give you \$10 on day 1, \$20 on day 2, \$30 on day 3, for all 30 days of a month. At the end of that month, what is the total amount your friend has given you?

3.11 Geometric Sequences

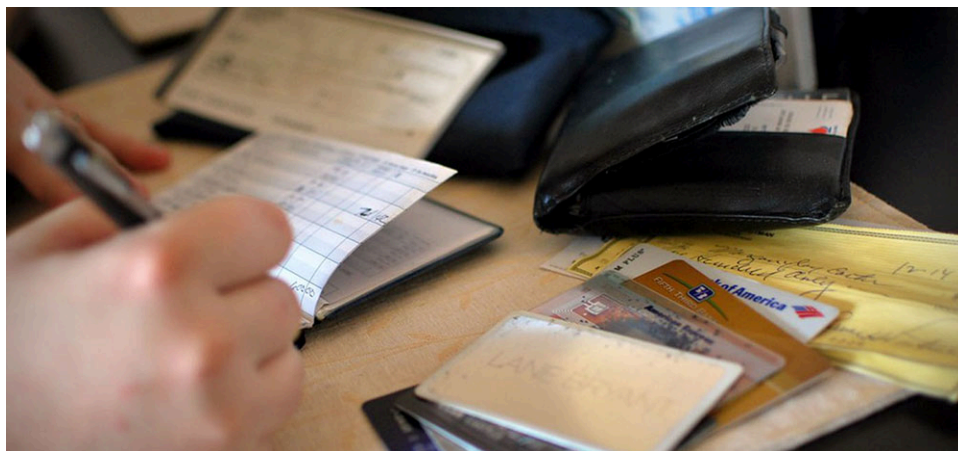


Figure 3.49 Savings grows in a geometric sequence. (credit: modification of “A big part of financial freedom is having your heart and mind free from worry about the what-ifs of life. – Suze Orman” by Morgan/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify geometric sequences.
2. Find a given term in a geometric sequence.
3. Find the n th term of a geometric sequence.
4. Find the sum of a finite geometric sequence.
5. Use geometric sequences to solve real-world applications.

One of the concerns when investing is the **doubling time**, which is length of time it takes for the value of the investment to be twice, or double, that of its starting value. A shorter doubling times means the investment gets bigger, sooner. For example, if you invest \$200 in an account with an 8-year doubling time, then in 8 years the value of the account will be double the starting amount, or $2 \times \$200 = \400 . After another 8 years (for a total of 16 years) the investment would be twice its value after the first 8 years, or $2 \times (2 \times \$400) = 2 \times (\$400) = \$800$. Every 8 years, the investment would double again, so after the third 8-year period, the investment would be worth $2 \times 2 \times (2 \times \$400) = \$1,600$. This process exhibits exponential growth, an application of geometric sequences, which is explored in this section.

Identifying Geometric Sequences

We know what a sequence is, but what makes a sequence a geometric sequence? In an arithmetic sequence, each term is the previous term plus the constant difference. So, you add a (possibly negative) number at each step. In a **geometric sequence**, though, each term is the previous term multiplied by the same specified value, called the **common ratio**. In the sequence $\{3, 6, 12, 24, 48, 96, 192, 384, 728, 1456\}$ the common ratio is 2. To see the difference between an arithmetic sequence and geometric sequence, examine these two sequences ([Figures 3.52](#) and [3.53](#)).

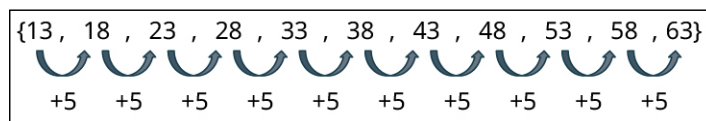


Figure 3.50 Arithmetic sequence

Each term in this arithmetic sequence is the previous term plus 5.

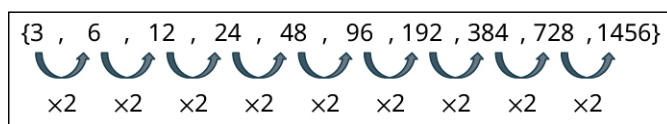


Figure 3.51 Geometric sequence

Each term in this geometric sequence is the previous term times 2.

In the sequence $\{3, 6, 12, 24, 48, 96, 192, 384, 728, 1456\}$, the numbers get big fairly quickly, and stay positive. However, that's not always the case with geometric sequences. Depending on the value of the common ratio, the terms could increase each time (like in the one shown in Figure 3.51), or the terms can get smaller each time, or the terms can alternate between positive and negative values. It all depends on the value of the common ratio, r .

Consider this geometric sequence:

$$\{5, 15, 45, 135, 405, 2025, \dots\}$$

Each term is the previous term times 5, which means the common ratio is 5. This common ratio is larger than 1, and so the terms increase each time. Now, look at this geometric sequence:

$$\{2, -6, 18, -54, 162, -486, 1458, \dots\}$$

Each term is the previous term times -3 , and the sign of the terms alternate from positive to negative. Then, there's this geometric sequence:

$$\left\{9, 3, 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots\right\}$$

Each term is the previous term times $\frac{1}{3}$, and the terms decrease each time. What we should take away from these three examples is if the common ratio is a positive number larger than 1, then the sequence increases. If the common ratio is a negative number, then the sign of the terms alternates between positive and negative. If the common ratio is between 0 and 1, then the terms decrease.

Two special cases of geometric sequences are when the constant ratio is 1 and when the common ratio is 0. When the constant ratio is 1, every term of the sequence is the same, as in $\{3, 3, 3, 3, 3, 3, 3, 3, 3\}$. This is referred to as a **constant sequence**. When the constant ratio is 0, the first term can be any number, but every term after the first term is 0, as in $\{-43.2, 0, 0, 0, 0, 0, 0, 0\}$.

EXAMPLE 3.138

Identifying Geometric Sequences

For each sequence, determine if the sequence is a geometric sequence. If so, identify the common ratio.

- $\{5, 20, 80, 320, 1,280, 5,120, 20,480, \dots\}$
- $\{-3, 6, -12, 24, 11, 33\}$
- $\{4, 2, 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots\}$

✓ Solution

- In the sequence $\{5, 20, 80, 320, 1,280, 5,120, 20,480, \dots\}$, the jump from 5 to 20 is a multiplication by 4, as is the next jump to 80, and the next to 320. Each term is 4 times the previous term. Since each term is 4 times the previous, this is a geometric sequence. The common ratio is 4.
- In the sequence $\{-3, 6, -12, 24, 11, 33\}$, notice that 6 is -3 times -2 . The jump from 6 to -12 is another multiplication by negative. So, if this is a geometric sequence, each term should be the previous term times -2 . But the change from 24 to 11 is not a multiplication by -2 . This means the sequence is not a geometric sequence.
- In the sequence $\{4, 2, 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots\}$, the change from 4 to 2 is a multiplication by $\frac{1}{2}$, as is the next jump, from 2 to 1, as is the next from 1 to $\frac{1}{2}$. Each term is $\frac{1}{2}$ times the previous term. Since each term is $\frac{1}{2}$ times the previous, this is a geometric sequence. The common ratio is $\frac{1}{2}$.

YOUR TURN 3.138

For each sequence, determine if the sequence is a geometric sequence. If so, identify the common ratio.

1. $\{-1, -5, -25, -125, -625, \dots\}$
2. $\{-3, 6, -12, 24, 11, 33\}$
3. $\{-500, 50, -5, \frac{1}{2}, -\frac{1}{20}, \dots\}$

As with arithmetic sequences, the first term of a geometric sequence is labeled a_1 . The number that is multiplied by each term is called the common ratio and is denoted r . So, if the first term is known, a_1 , and the common ratio is known, r , then the n th term, a_n , can be calculated with the formula $a_n = a_1 r^{n-1}$.

FORMULA

The n th term of the geometric sequence, a_n , with first term a_1 and common ratio r , is $a_n = a_1 r^{n-1}$.

Return to the sequence $\{3, 6, 12, 24, 48, 96, 192, 384, 728, \dots\}$. We observe that the first term is 3, so $a_1 = 3$. We also found that the common ratio is 2, so $r = 2$. The table below shows how any term can be calculated using just a_1 and r .

i , Place in Sequence	a_i , i^{th} Term	Value of Term	Term Written as $a_1 \times r^{i-1}$
1	a_1	3	3×2^0
2	a_2	6	3×2^1
3	a_3	12	3×2^2
4	a_4	24	3×2^3
5	a_5	48	3×2^4
i	a_i		$3 \times 2^{i-1}$

EXAMPLE 3.139

Determining the Value of a Specific Term in a Geometric Sequence

In the following geometric sequences, determine the indicated term of the geometric sequence with a given first term and common ratio.

1. Determine the 9th term of the geometric sequence with $a_1 \times 6$ and $r = 3$.
2. Determine the 11th term of the geometric sequence with $a_1 = 2$ and $r = -5$.

Solution

1. Using $a_n = a_1 r^{n-1}$ with $a_1 = 6$, $r = 3$, and $n = 9$, we calculate

$$a_9 = a_1 r^{9-1} = 6 \times (3)^{9-1} = 6 \times (3)^8 = 6 \times 6561 = 39366$$

The 9th term of the geometric sequence with $a_1 = 6$ and $r = 3$ is $a_9 = 39366$.

2. Using $a_n = a_1 r^{n-1}$ with $a_1 = 2$, $r = -5$, and $n = 11$, we calculate

$$a_{11} = a_1 r^{11-1} = 2 \times (-5)^{11-1} = 2 \times (-5)^{10} = 2 \times 9,765,625 = 19,531,250$$

YOUR TURN 3.139

In the following geometric sequences, determine the indicated term of the geometric sequence with a given first term and common ratio.

1. Determine the 12th term of the geometric sequence with $a_1 = 3072$ and $r = \frac{1}{2}$.
2. Determine the 5th term of the geometric sequence with $a_1 = 0.5$ and $r = 8$.

VIDEO

[Geometric Sequences \(https://openstax.org/r/Geometric_Sequences\)](https://openstax.org/r/Geometric_Sequences)

Finding the Sum of a Finite Geometric Sequence

As with arithmetic sequences, it is possible to add the terms of the geometric sequence. Like arithmetic sequences, the formula for the finite sum of the terms of a geometric sequence has a straightforward formula.

FORMULA

The sum of the first n terms of a finite geometric sequence, written s_n , with first term a_1 and common ratio r , is $s_n = a_1 \left(\frac{1-r^n}{1-r} \right)$ provided that $r \neq 1$.

EXAMPLE 3.140

Calculating the Sum of a Finite Geometric Sequence

1. What is the sum of the first 13 terms of the geometric sequence with first term $a_1 = 5$ and common ratio $r = 3$?
2. What is the sum of the first 7 terms of the geometric sequence with first term $a_1 = 16$ and common ratio $r = \frac{1}{8}$?

Solution

1. Using $a_1 = 5$, $r = 3$, and $n = 13$, we find that the sum is:

$$S_{13} = a_1 \left(\frac{1-r^{13}}{1-r} \right) = 5 \left(\frac{1-3^{13}}{1-3} \right) = 5 \left(\frac{1-1,594,323}{-2} \right) \\ = 5 \left(\frac{-1,594,322}{-2} \right) = 5 (797,161) = 3,985,805$$

The sum of the first 13 terms of this geometric sequence is 3,985,805.

2. Using $a_1 = 16$, $r = \frac{1}{8}$, and $n = 7$, we find that the sum is:

$$s_7 = a_1 \left(\frac{1-r^7}{1-r} \right) = 16 \left(\frac{1-\left(\frac{1}{8}\right)^7}{1-\left(\frac{1}{8}\right)} \right) = 16 \left(\frac{1-\left(\frac{1}{2,097,152}\right)}{\frac{7}{8}} \right) \\ = 16 \left(\frac{2,097,151/2,097,152}{7/8} \right) = 16 \left(\frac{299,593}{262,144} \right) = \frac{299,593}{16,384} = 18.2857$$

The sum of the first 7 terms of this geometric sequence is 18.2857.

YOUR TURN 3.140

1. What is the sum of the first 10 terms of the geometric sequence with first term $a_1 = 7$ and common ratio $r = 6$?
2. What is the sum of the first 6 terms of the geometric sequence with first term $a_1 = 27$ and common ratio $r = \frac{1}{3}$?

Using Geometric Sequences to Solve Real-World Applications

Geometric sequences have a multitude of applications, one of which is compound interest. Compound interest is something that happens to money deposited into an account, be it savings or an individual retirement account, or IRA.

The interest on the account is calculated and added to the account at regular intervals. This means the interest that was earned later gains its own interest. This allows the money to grow faster. If that interest is added every month, we say it is compounded monthly. If the interest is added daily, then we say it is compounded daily. The amount of money that is deposited into the account is called the principal and is denoted P . The account earns money on that principal. The amount it earns is a percentage of the money in the account. The interest rate, expressed as a decimal, is denoted r .

FORMULA

If you deposit P dollars in an account that earns interest compounded yearly, then the amount in the account, A , after t years is calculated with the formula: $A = P(1 + r)^t$. This is a geometric sequence, with constant ratio $(1 + r)$ and first term $a_1 = P$.

EXAMPLE 3.141

Calculating Interest Compounded Yearly

Daryl deposits \$1,000 in an account earning 4% interest compounded yearly. How much money is in the account after 25 years?

Solution

Using $A = P(1 + r)^t$ with $P = 1000$, $r = 0.04$, and $t = 25$, we find that

$A = P(1 + r)^t = 1,000 \times (1 + 0.04)^{25} = 1,000 \times (1.04)^{25} = 1,000 \times 2.66583633 = 2,665.85$. After 25 years, there is \$2,665.84 in the account.

YOUR TURN 3.141

1. Sophia deposited \$4,000 in an account that earns 5.5% interest compounded yearly. After 20 years, Sophia withdrew all the money in the account to pay for her child's college. How much money was in the account when Sophia withdrew the money?

Another application of geometric sequences is exponential growth. This arises in biology quite frequently, especially in relation to bacterial cultures, but also with other organism population models. In bacterial cultures, the time it takes the population to double is often recorded. This time to double is the same, regardless of how big the population gets. So, if the population doubles after 3 hours, it doubles again after another 3 hours, and again after another 3 hours, and so on. Put into geometric sequence language, it has a common ratio of 2.

EXAMPLE 3.142

Doubling a Bacterial Culture

When *Escherichia coli* (*E. coli*) is in a broth culture at 37°C, the population of *E. coli* doubles in number with 30 organisms, how many *E. coli* bacteria are present in the culture after 16 hours?

Solution

Since the population is doubling every 20 minutes, this is a geometric sequence situation with common ratio $r = 2$. The culture begins with 30 organisms, so $a_1 = 30$. The time, 16 hours, is 48 twenty-minute periods, so we're looking for the 48th term in the sequence. Using these values in the geometric sequence formula gives

$$a_{48} = a_1 r^{n-1} = 30 \times 2^{48-1} = 30 \times 2^{47} = 30 \times (1.40737 \times 10^{14}) = 4.22212 \times 10^{15}.$$

So, after 16 hours, the culture contains 4.22212×10^{15} *E. coli* organisms. That's more than 4,000 trillion bacteria.

YOUR TURN 3.142

1. When *Streptococcus lactis* (*S. lactis*) is in a milk culture at 37°C, the population of *S. lactis* doubles in number every 30 minutes. If the culture began with 15 organisms, how many *S. lactis* bacteria are present in the culture

after 20 hours?

EXAMPLE 3.143

Applying the Sum of a Finite Geometric Sequence

A player places one grain of rice on the first square of a chess board. On the second square, the player places 2 grains of rice. On the third square, the player places 4 grains of rice. On each successive square of the board, the player doubles the number of grains of rice placed on the chess board. When the player places the last rice on the 64th square, how many total grains of rice have been placed on the board?

Solution

Since the number of grains of rice is doubled at each step, this is a geometric sequence with first term $a_1 = 1$ and common ratio $r = 2$. Rice is placed on 64 total squares, so we want the sum of the first 64 terms. Using this information and the formula, the total number of grains of rice on the board will be:

$$\begin{aligned} s_{64} &= a_1 \left(\frac{1-r^{n+1}}{1-r} \right) = 1 \times \left(\frac{1-2^{64+1}}{1-2} \right) = \left(\frac{1-2^{65}}{-1} \right) = -(1-2^{65}) \\ &= -(-9.2233720369 \times 10^{18}) = 9.2233720369 \times 10^{18} \end{aligned}$$

That's a 20-digit number!

YOUR TURN 3.143

1. You have a square 1 meter on each side. You begin by coloring one half of the square blue. Then you color half the remaining area blue. Then you color half the remaining area blue once more. At each step, you color half the remaining area. What is the total area you have colored blue after performing this process 15 times?

VIDEO

[Sum of a Finite Geometric Sequence \(https://openstax.org/r/Sum_of_a_Finite_Geometric_Sequence\)](https://openstax.org/r/Sum_of_a_Finite_Geometric_Sequence)

Check Your Understanding

58. Is the following a geometric sequence? Explain.
 $\{3, 6, 12, 24, 48, 96, 192\}$
59. Find the common ratio of the geometric sequence $\{3, -30, 300, -3,000, \dots\}$.
60. In a geometric sequence, the first term is 10 and the common ratio is 1.5. What is the 15th term?
61. What is the sum of the first 100 terms of the geometric sequence with first term 4 and common ratio 0.3?
62. \$15,000 is deposited in an account the yields 4.2% interest compounded annually. How much is in the account after 17 years?



SECTION 3.11 EXERCISES

For the following exercises, determine if the sequence is a geometric sequence.

1. $\{3, 7, 11, 15, 25, 100, \dots\}$
2. $\{2, 4, 8, 16, 32, \dots\}$
3. $\{9, 0.9, 0.09, 0.009, 0.00009, \dots\}$
4. $\{262,144, 65,536, 16,384, 4,096, 1,024, \dots\}$
5. $\{14, 19, 24, 29, 34, 50, 60\}$
6. $\{3.9, 2.3, 0.7, -0.9, -2.5, -4.1, -5.7, \dots\}$
7. $\{4, -8, 16, -32, 64, -128, 256, \dots\}$
8. $\{8, -4, 2, -1, 0.5, -0.25, 0.125, -0.0625, \dots\}$

For the following exercises, the sequences given are geometric sequences. Determine the common ratio for each. Verify that each term is the previous term times the common ratio.

9. $\{3, 6, 12, 24, 48, 96, \dots\}$
10. $\{8, 24, 72, 216, 648, 1944, \dots\}$
11. $\{15, 3, 0.6, 0.12, 0.024, 0.0048, 0.00096, \dots\}$
12. $\{52, 26, 13, 6.5, 3.25, 1.625, 0.8125, 0.40625, \dots\}$
13. $\{18, -18, 18, -18, 18, -18, \dots\}$
14. $\{48, -12, 3, -0.75, 0.1875, -0.046875, \dots\}$

For the following exercises, the first term and the common ratio of a geometric sequence is given. Using that information, determine the indicated term of the sequence.

15. $a_1 = 5, r = 3$, find a_6 .
16. $b_1 = 7, r = 9$, find b_5 .
17. $c_1 = 11, r = 4$, find c_{12} .
18. $a_1 = 2, r = 7$, find a_9 .
19. $t_1 = 100, r = \frac{1}{5}$, find t_{10} .
20. $b_1 = 56, r = 0.25$, find b_{15} .
21. $b_1 = 13, r = -2$, find b_{10} .
22. $a_1 = 11, r = -3$, find a_{12} .
23. $a_1 = 12, r = -\frac{1}{3}$, find a_8 .
24. $a_1 = 100, r = -10$, find a_{15} .

For the following exercises, the first term and the common ratio is given for a geometric sequence. Use that information to find the sum of the first n terms of the sequence, s_n .

25. $a_1 = 3, r = 4$, calculate s_5 .
26. $a_1 = 5, r = 3$, calculate s_9 .
27. $a_1 = 4, r = 5$, calculate s_8 .
28. $a_1 = 48, r = 2$, calculate s_{11} .
29. $a_1 = 450, r = 0.5$, calculate s_{12} .
30. $a_1 = 300, r = 0.25$, calculate s_{10} .
31. $a_1 = 3, r = -2$, calculate s_{11} .
32. $a_1 = 5, r = -4$, calculate s_8 .

For the following exercises, apply your understanding of geometric sequences to real-world applications.

33. *Lactobacillus acidophilus* (*L. acidophilus*) is a bacterium that grows in milk. In optimal conditions, its population doubles every 26 minutes. If a culture starts with 20 *L. acidophilus* bacteria, how many bacteria will there be after 390 minutes? Hint: This means the 26-minute time period has occurred 15 times.
34. *Bacillus megaterium* (*B. megaterium*) is a bacterium that grows in sucrose salts. In optimal conditions, its population doubles every 25 minutes. If a culture starts with 30 *B. megaterium* bacteria, how many bacteria will there be after 1,000 minutes? Hint: This means the 25-minute time period has occurred 40 times.
35. Alex and Jill deposit \$4,000 in an account bearing 5% interest compounded yearly. If they do not deposit any more money in that account, how much will it be worth in 30 years?
36. Kerry and Megan deposit \$6,000 dollars in an account bearing 4% compounded yearly. If they do not deposit any more money in that account, how much will be in the account after 40 years?
37. You decide to color a square that measures 1 m on each side in a very particular manner. You first cut the square in half vertically. You color one side of the square with purple. On the side of the square that was not colored, you draw a line dividing that region horizontally exactly in half. You color the lower half blue. Now, you cut the remaining quarter of the square precisely in half with a vertical line. You color the left side red. You repeat this process 12 times. After you color that 12th piece, what is the total area you have colored?
38. Consider the geometric sequence with first term 0.9 and common ratio of 0.1. What is the sum of the first 5 terms?
39. Repeat Exercise 38, for the sum of the first 10 terms.

For the following questions, recall that the formula for interest compounded yearly is $A = P(1 + r)^t$, where A is the amount in the account after t years, P is the initial amount deposited, and r is the interest rate per year. However, if the account is compounded monthly, the formula changes to $A = P\left(1 + \frac{r}{12}\right)^{12t}$.

40. Returning to Kerry and Megan (Exercise 36), what would their account be worth if their account was compounded monthly?
41. Returning to Alex and Jill (Exercise 35), what would their account be worth if their account was compounded monthly?
42. Imagine your family tree. You have two parents. Your parents have two parents: your grandparents. And so on.

How many great-great-great-great-grandparents do you have? Hint: This would be six generations back.

- 43.** Imagine your family tree. You have two parents. Your parents have two parents: your grandparents. And so on. How many great (20 times) grandparents do you have? Hint: This would be 22 generations back.